Media Capture & Playback Validation – Practical Rules

1. Never trust a single player

If it plays in only one player, it is not validated. Different players use different demuxers, codecs, clocks, and buffering strategies.

2. Always test with two validation paths

• Independent player (e.g. VLC) – validates standards compliance

• Stack-native player (MfPack Media Foundation samples) – validates MF correctness

3. Default system players are not reference implementations

Windows 11 Media Player may mis-handle valid files, uncommon codecs, or timing. Failure here does not automatically mean your file is broken.

4. Same API ≠ same behavior

Two applications using Media Foundation can behave very differently due to topology, clocking, buffering, and renderer choices.

5. Isolate the pipeline when debugging

Test video-only, audio-only, and muxing separately. Never debug capture and playback at the same time.

6. Control your test environment

Reduce variables. Use VLC and MfPack player samples instead of unreliable defaults.

7. Log what you already know

At capture time you already know codecs, resolution, FPS, sample rate, channels, and bitrates. Log them.

Professional rule:

If a file plays correctly in VLC *and* in MfPack Media Foundation samples,
the file is correct — even if a default system player disagrees.